

VISIONCLAW AGENT PLATFORM — COMPREHENSIVE FEATURES

AI Buddy LLC (Illinois) | April 2026

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PLATFORM STATS

- 130+ server-side TypeScript files (~89,000 lines)
 - 38+ frontend pages | 66 database tables
- 96+ built-in AI tools + custom tool support
 - 23 active skills | 14 AI personas
 - 36+ models across 8+ providers
- 40 governance rules | 11 research programs
- 7 research schedules | 13 heartbeat tasks

AUTHENTICATION & SECURITY

- Multi-method: Replit Auth, Email+Password, Admin PIN (HMAC-SHA256)
 - DB-backed sessions with timing-safe cryptography
 - Password policy enforcement, email verification
- Helmet CSP, provider key proxy, IronClaw SafetyLayer
 - Per-tool rate limiting (sliding window per tenant)

MULTI-TENANCY

- Strict tenant isolation across all data
- Per-tenant memory backup, file storage, billing
- Admin (tenant 1) with PIN-gated admin endpoints
 - BYOK tier system for API key management

CHAT ENGINE & AI ROUTING

- Real-time AI responses via SSE
- Smart Model Auto-Selection with Task Complexity Classifier
 - Multimodal-Aware Routing, Auto-Thinking Mode
- Subscription-First Routing (BYOS): OAuth tokens as PRIMARY inference
- Claude Runner Bridge: \$0 per-token Anthropic routing via Max plan
 - Cost-aware auto-routing: free > cheap > paid

- All 36 models tagged with costClass

14 AI PERSONAS

1. VisionClaw (CEO) — Platform orchestrator
2. Felix — Multi-step task decomposition & orchestration
3. Forge — Code generation & technical execution
4. Teagan — Teaching & education
5. Blueprint — Architecture & design
6. Chief of Staff — Operations & stability
7. Scribe — Documentation & writing
8. Proof — Quality assurance & adversarial verification
9. Radar — Research & intelligence
10. Neptune — Data analysis & visualization
11. Apollo — Creative content & media
12. Atlas — Project management
13. Cassandra — Finance & market intelligence
14. Luna — Personal assistant & scheduling

96+ AITOLS

**Communication | Research & Web | Memory & Knowledge |
Documents & Files**

**Voice & Audio | Code & Execution | Agentic Operations | Project
Management**

Governance | System | Payments | Visualization

INTELLIGENCE & MEMORY

- Hierarchical Memory Graph, Three-Tier Semantic Memory
 - BM25/Vector/Hybrid Document Search
 - Zero-Loss Compaction, pgvector similarity
 - DreamTask Memory Consolidation (6h cycle)
- LLM-Judged Memory Relevance Scoring (Claude Code inspired)
 - GPT-4.1 Mini selects best 5-7 from 15-20 candidates
 - Smart filtering: skips generic docs, keeps warnings/gotchas
 - 2s timeout with vector fallback, 30s context-aware cache
 - Instinct Learning & Proactive Context Loading

AGENTIC INFRASTRUCTURE

- Persistent Agent Desks, Internal Channels, Event Bus
 - Multi-Layer Delegation (up to 5 levels)
- Felix Auto-Orchestration (DAG decomposition)
 - Execution Supervisor with circuit breaker
 - Auto-QA Pipeline (Proof reviews outputs)
 - Per-Task Cost Tracking with live badges
 - Adversarial Verification Agent

ENHANCED STUCK DETECTION (Claude Code Inspired)

- Circular loop detection with deferred message injection
 - Stalled delegation tracking (60s periodic check)
- Hung HTTP detection with per-tool timeouts + 5-min dedup
 - Admin endpoint: GET /api/admin/diagnostics/stuck

PROCESS GOVERNOR (40 Rules, 7 Categories)

AGENCY EXPANSION FRAMEWORK (6 Tiers)

AUTONOMOUS OPERATIONS (Heartbeat, Watchdog, Self-Tuning)

NIGHTLY AUTORESEARCH (11 Programs, Adaptive Backpressure)

SKILLIFY — Session-to-Skill Extraction

AGENT LIVE STATUS SUMMARIES

PRESENTATION & VISUALIZATION

- Mermaid diagrams, Charts (Recharts), Dashboards
 - AI images, Video production, PowerPoint slides
- VibeVoice ASR, Live delegation feed with voice narration

EXTERNAL INTEGRATIONS

AI: OpenAI, Anthropic, Gemini, xAI, Perplexity, OpenRouter

Payments: Stripe, Coinbase | Services: ElevenLabs, Firecrawl

Communications: AgentMail, WhatsApp, Discord, Telegram

Storage: PostgreSQL + pgvector, Replit Object Storage

TECH STACK

Backend: Express.js, TypeScript, Drizzle ORM, Zod, Helmet

Frontend: React 18, Vite, shadcn/ui, TailwindCSS, TanStack Query

v5

Database: PostgreSQL 66 tables, pgvector, Drizzle ORM